THESIS MANUSCRIPTS

For BSCS

Thesis General Format

The following are the parts of the study, optional and necessary, that must be followed and must appear in this order.

Title Page.

This page shows the thesis title, the names of the students who wrote it (called proponents), and a note about the degree or qualification they're earning with this thesis. It also lists the name of the school or university where the thesis is being submitted, plus the month and year when it's handed in.

Approval Sheet.

This page is called the Approval Sheet. It lists the names of the student(s) who did the thesis (called proponent/s) and the thesis title. It also has the signatures of the adviser, the college dean, and the members of the oral defense panel. This page proves that the thesis has been officially approved, and it must include the date when the approval happened.

Acknowledgements.

This part thanks the people and groups who helped the students finish their thesis. Keep the thanks simple and polite.

Abstract.

This is a short summary of the whole thesis. It covers the problem the study looked at, the goals, the methods used, the main results, why it matters, and the final thoughts. The abstract needs at least 200 words but no more than 500 words. Type it with single spacing. Usually, it does not mention other books or studies..

Table of Contents.

This is a list of all the main parts of the thesis, in order, with their page numbers. It includes chapter titles, section titles, subsection titles, the references list, and appendices. It also lists the titles of the front pages and required forms.

List of Tables.

A comprehensive listing of titles of all tables found in the body of the thesis with indication of the corresponding page numbers. Tables should be numbered in sequence, using Arabic numerals.

List of Figures.

List of charts, graphs, maps and other illustrations used in the thesis. Specific items consisting of 12 or

more must be listed separately. For example, if there are more than 12 illustrations, then there must be a separate list of illustrations. List the exact title or caption of each figure and its corresponding page. Figures should also be numbered in sequence, using Arabic numerals.

List of Appendices.

An appendix or appendices, if any, should be after the References. Appendices include original data, preliminary tests, tabulations, tables that contain data of lesser importance, very lengthy quotations, forms and documents, computer printouts and other pertinent documents. Appendices should be arranged chronologically as they are cited in the main text. Use capital letters of the English alphabet to track appendices, and always begin with the letter A.

List of Abbreviations and Symbols (Optional).

All abbreviations used in the thesis are listed for easy reference of the reader. This section, however, is optional.

Definition of Terms

This section explains important words used in the thesis. It gives clear meanings based on how they are used in this study. This helps readers understand the problem better and avoids confusion from words that could mean different things.

Operational Terms: These explain words as they are used in this specific study or field. List them in alphabetical order.

Technical Terms: These are words with standard meanings that most people already know. Get their definitions from dictionaries or books, and list them in alphabetical order.

THESIS CONTENT.

The main body of a thesis is divided into five (5) chapters, which may be further subdivided into sections.

Chapter 1 – Introduction and Its Background

Discuss about the current state of technology and the topic's situation today, both locally (in your area) and internationally (around the world), as it relates to your chosen subject. This part explains why the researchers did this work and gives enough background to show why the study matters and is important.

This section starts the research paper with a quick overview of the project. A good introduction covers some or all of these: (a) the problems being studied, (b) an unhappy situation or need that requires a fix to close a technology gap, (c) reasons and proof for picking this topic, (d) the history or background of the problem.

Researchers must explain why the new software is needed. It should match the technology gap they found during their analysis. Most importantly, describe the algorithm(s) (step-by-step plans) that the software will use, and include them in the solutions discussion.

Research Problem

A research problem is a clear statement about something worrying, like a bad situation that needs fixing, a hard challenge to remove, or a tough question from books, theories, or real life. It shows why we need a deep look and careful study.

This part also includes the main (general) problem that the new system will solve. It lists 3 to 5 specific problems that the system will fix once it's done.

Research Objective

This part has a main (general) goal that says what the study wants to achieve overall. It also lists specific goals that the researchers will do to reach the main goal. These goals must match the problems above and be ones we can actually answer.

Research Framework

This section explains the key theory or theories that guide the researchers while designing or building the thesis. These theories provide a strong scientific base or support for the study, drawn from the Review of Literature. Use two frameworks for the study: (1) The theoretical framework, shown clearly in how algorithms will work in software development. (2) The conceptual framework, made by researchers from the theoretical one, draw or illustrate it. A conceptual framework mixes ideas from past studies to explain something new. It maps the steps for your study, using knowledge from other researchers and your own views. Combine variables from articles you've read, guided by your problem statement. Your work answers a question others haven't, filling a knowledge gap.

Scope and Limitations of the Research

The scope and limitation include the brief statement of the general purpose of the study and the subject matter and topic studies discussed. Also, the population or universe from which the respondents are selected. This must be large enough to make the generalization significant. In the limitation of the study, this section must include the weaknesses of the study that is beyond the control of the researchers.

The scope identifies the coverage of the study in terms of subject, facilities, objectives, area, time duration of the issues to which the research is focused.

The limitation of the study defines the constraints or weaknesses which are not within the control of the writer, therefore they are not expected to be covered by the study.

Significance of the Research

This section explains why the research focuses on this area, what new contributions it will make, and who will benefit. List the beneficiaries from biggest to smallest groups (macro to micro), like from society or country down to specific people or schools.

Chapter 2 – Review of Literatures

This section explains the features, strengths, and limits of existing studies, algorithms, theoretical frameworks, and software applications that are relevant or very similar to your proposed topic. You can organize your review in order of time (chronological) or by topic (thematic). At the end, include a short and clear synthesis that repeats the main findings from the literature you reviewed. Summarize the content of the gathered studies and highlight information that directly supports your research. You may also mention and discuss similar apps or systems to show why your study matters.

Synthesis

This part of the Review of Literature shows how your current study relates to the research you reviewed. It summarizes and stresses why your study is important. If it fits your study, you can include a table called an algorithm matrix to support your choice of algorithm. This table helps compare algorithms and shows why you picked the one you used.

Chapter 3- Research Methodology

Materials and Methods is the chronological listing of methods, procedure and material used by the proponent/s. Methods used for gathering of data, laboratory and field experiment, theoretical and/or conceptual frameworks.

Research Design

This section explains the research design, which is the overall plan you choose to connect all parts of your study in a clear and logical way. This plan makes sure you can effectively solve your research problem. It serves as a blueprint for collecting, measuring, and analyzing data. Remember: your research problem decides which design you should use, not the other way around. For computer science, the research design can combine an Experimental study (testing different algorithms to find the one that works best) with Development and Application studies (to prove the concept and build a software prototype).

Locale of the Study

This section discusses the target locale for this research. Also, the nature (who, where, what) of the selected locale where the researchers will conduct a research study, this should be based on the objectives to be achieved.

Applied Concepts and Techniques

This section lists the concepts and techniques used to solve the research problem and reach the research goals. Mention theoretical computer science fields like algorithms, theory of computation, information theory, and automation, or data-driven fields like data science, data mining, deep learning, machine learning, or Artificial Intelligence. Clearly explain the theoretical background that matches your research focus. For example: Machine Learning → Unsupervised Learning → Clustering → K Means Clustering / hierarchical clustering / Density-Based Spatial Clustering of Applications with Noise (DBSCAN) / Gaussian Mixtures Model.

Algorithm Analysis

This section describes the methods or techniques used to analyze data or algorithms. Algorithm analysis is a way to measure how well an algorithm performs. Speed is one of the most important factors in judging an algorithm's potential. Talk about the complexity and the metrics used to measure performance. Other important factors include performance, accuracy, time complexity, space complexity, and other quality measures. This section must also cover the required criteria for an algorithm: input, output, clear steps (definiteness), must end (finiteness), and effectiveness (it should get the job done). For example, in machine learning or deep learning models, performance is checked using: confusion matrix, accuracy, precision, recall, specificity, F1 score, Precision-Recall curve, or Receiver Operating Characteristics (ROC) curve.

Data Collection Methods

This section states the data source and nature of the data, present the data sets used in a table with its

attribute, description, and values.

Data Model Generation

This section states the framework in generating data model as stated on the objectives of the study. Elaborate the phases of the selected CS solutions. In data mining, CRISP-DM to KDD is usually used enumerate the phases and discuss your techniques in achieving every tasks. In ML Process there are several frameworks which can be generally broken into several steps: such as Collection of Data, Data Wrangling, Model Building, Model Evaluation and Model Deployment. Please ensure that all methods, techniques, and approach will be based on scientific process and reputable references.

System Development Methodology

This section discusses the researchers' developmental methodology as the algorithm or data model integration in the software design. The models specify the various stages of the process and the order in which they are carried out. Software process model may also be reflected on this section.

Software Tools Used

This section discusses the programming language and software tools used in data modelling and system development. Elaborate the specific tasks and uses of each tool.

System Architecture

This section presents the system architecture of the developed system. Enumerate the functions and features highlight the integration of computing science solutions. Discuss the structure, behavior, and more views of a system.

Software Testing

This section discusses the software testing method and strategies to determine if the computing science problem was solved.

Chapter 4 - Results and Discussion

This chapter shows the results and discussion that support the methods used to reach the study's goals. The presentation is usually organized by the specific objectives or research problems. This chapter also presents the algorithm and model created by the researchers. Explain in detail the algorithm you designed and developed, including how it will be applied in your study and how well it fits your topic.

This section explains how the algorithm is used in building the software. It shows the pseudo code (a step-by-step plain-language description) of how the algorithm works in your study, with a detailed explanation of each part. It also presents and discusses the results from running the algorithm or from data analysis.

Chapter 5 – Summary, Conclusions, and Recommendations

This chapter includes a summary of the main findings of the study. It also presents the significance of the study and relates findings to the objectives and problems written in the introduction part of the study. Recommendation/s must be stated in this chapter. This part usually directs the reader to conduct further research on some specific areas related to the study.

This chapter includes a summary of the main findings of the study. It also presents the significance of the study, and relates findings to the objectives and problems written in the introduction part of the study. Recommendation/s must be stated in this chapter. This part usually directs the reader to conduct further research on some specific areas related to the study.

Summary

This part briefly summarizes all the important results of your research. List and review all key ideas without critiquing them. The goal is to clearly show the main points of your study to the reader.

Conclusions

This section presents conclusions based on the study's objectives and the findings you discovered. It merges what you set out to do with what you actually found.

Recommendation

This part gives recommendations based on your conclusions. It may suggest directions for future research or explain how to use the software product to get the most benefit.

Literature Cited. This is a list of works cited, as well as works consulted but not cited (example, background reading not necessarily cited) in the construction of the research paper. The list of references is numbered and arranged alphabetically. For format on the writing of references, the APA (American Psychological Association) style of documentation shall be followed.

APPENDICES

A. Technical Background *(diagrams may vary as per applicability)*

a. Planning & Requirement Analysis Phase
   - Preliminary Investigation Answered Questionnaires
   - Transcribed Interview, Observations
   - Story Board
   - Current Flow Chart / Process Flow
   - Proposed Flow Chart / Process Flow
   - Proposed Use Case Diagram

b. Specification & Design Phase
   - System Architecture / Module Specification
   - System Context Diagram
   - Data Flow Diagram
   - Data Dictionary
   - Hierarchical Input Process Output Model

c. Project Schedule
d. Hardware and Software Resources
e. Input/Output/Reports Screen Shots
f. Testing & Evaluation Instruments
g. Implementation Plan
h. User's Manual

B. **Communication Letters & Forms** – all approved letter, including the letter if the considerations, changing of titles, software, and version. Consultants and technical adviser's approval and recommendation letters.
   - Request Letter
   - Pre-Proposal Approval
   - Proposal Approval

- Recommendation for Final Oral Defense
- ISO Forms

C. **Certificate of Acceptance / MOA / MOU** – Certificate with Proper Documentation, MOA & MOU for Technology Transfer

D. Curriculum Vitae of the Researchers

E. Other Certifications

FOR TEXTUAL FORMAT

1. Manuscripts to be submitted should be complete with preliminary pages (Title Page, Approval Page, Acknowledgement, Table of Contents, List of Figures, List of Appendices and Abstract);
2. The preliminary pages should be included in the Table of Contents with corresponding small Roman numerals as page nos.;
3. In the approval page, author, thesis titles, degree and major field should be all capitalized.
4. ONLY ONE PAGE is allowed for the Acknowledgement, can be presented in single space;
5. Abstract should not be more than 200 words;
6. The main headings (i.e., CHAPTER I – INTRODUCTION, CHAPTER II – REVIEW OF RELATED LITERARTURE, CHAPTER III – RESEARCH METHODOLOGY, CHAPTER IV – RESULTS AND DISCUSSION, CHAPTER V – SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS, and LITERATURE CITED) should be all capitalized and should have imaginary page numbers. These headings should be at the center of the page and after 1 inch from the top side;
7. The page no. 2 and the succeeding page numbers should be placed 1 inch from the top and 1 inch from the right side at the upper right-hand corner of the page numbers and should be written in small Arabic numbers;
8. Maintain 4 spaces between the main heading and the succeeding major subsection;
9. Major subsection is place at the left side of the page with first letter capitalized and in inverted pyramid form when composed of five or more words;
10. Have 3 spaces between the major subsection and minor subsection;
11. Minor subsection is placed at the left side of the page, with first letters capitalized and when it runs more than half the page, it should be cut off, flush to the left margin in single space;
12. Have 3 spaces between the minor subsection and paragraph heading;
13. Paragraph heading should be indented five spaces to the right, with only the beginning letter is capitalized, italicized and with paragraph immediately follows after two spaces.
14. Maintain only 3 levels of subsections (major, minor, subsections and paragraph headings);
15. Do not underscore when defining terms and for emphasizing ideas;
16. Always use double spaces in the text and single space for tables and figures;
17. Use Times New Roman, size 12 for font style and font size, respectively;
18. Margins should be 1inch top, bottom and right, while 1.5 inches left;
19. For table of contents, Subsections should be included and should be presented using indents and Tab, main headings and Preliminary page titles should be presented capitalized and bold

FOR TABULAR PRESENTATION:

1. Table could be placed along with the text, maintain 4 spaces before and after the text/paragraphs;
2. Tables should be presented with double solid lines on its top and bottom parts;
3. First level heading should be all capitalized and in bold format;
4. Second level heading should have first letters capitalized;
5. Third level heading should have only small letters;
6. Tables should not have side boxes
7. For continued tables, there is no need to indicate table title, instead use Table _continued… However, indicate table heading with double solid lines at the top and until the bottom part of the table;
8. Cut too many lines within the tables, if possible only vertical lines to separate the headings;
9. Table title should be placed on top of the table and it should be 1 space before the double solid lines;
10. Tables can be presented after it is mentioned in the text or it could be placed as appendix tables.

Table 1. Proximate nutritional composition of dried azolla and duckweed.

% COMPOSITION	AZOLLA	DUCKWEED
Dry Matter	9.4	9.1
Ash Content	16.1	21.5
Crude Protein	20.9	17.1
Crude Fiber	22.3	16.3
Ether Extract	1.66	0.9
Nitrogen Free Extract	29.6	35.2

FOR FIGURES PRESENTATION:

1. Figures should be clear and do not contain unnecessary marks;
2. Figures caption should be placed 2 spaces after the figure is presented;
3. Figures could also be indicated along with the text or in separate page. If placed with the text, maintain spaces before and after the text;
4. Figures can be presented after it is mentioned in the text or could be placed as appendix figures;
5. Do not enclose the figure in box

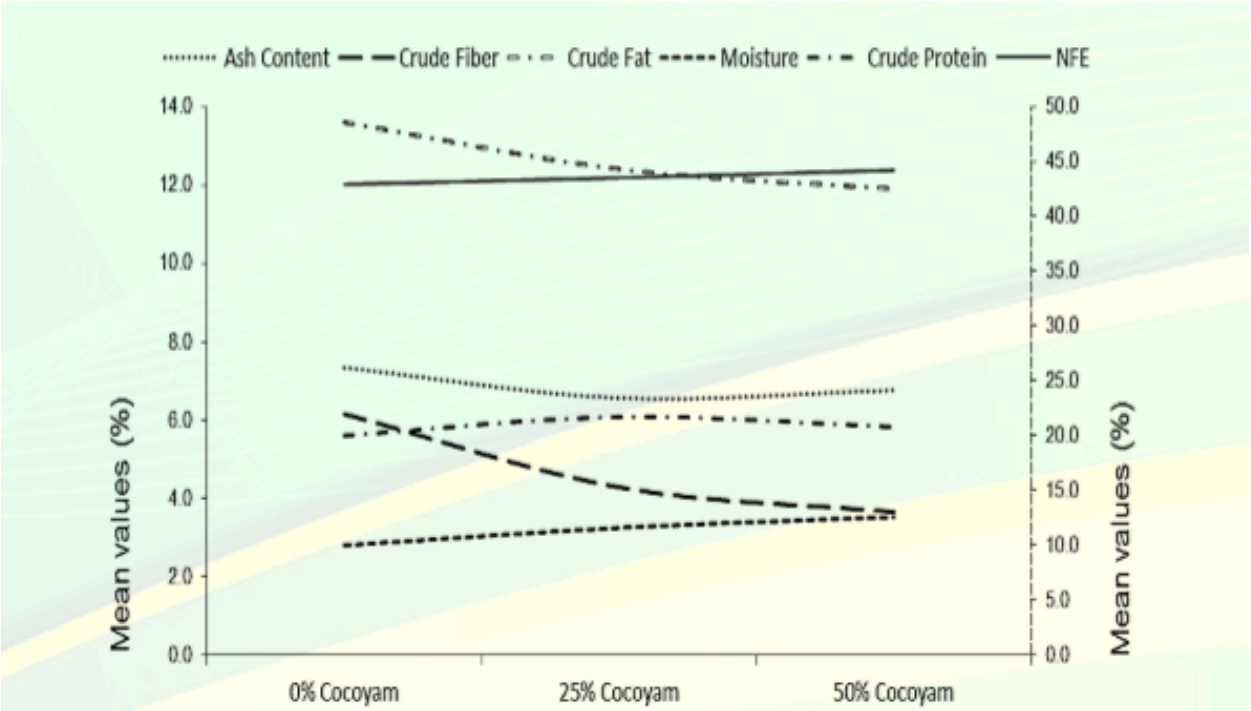


Figure 1. Trends in proximate values for moisture, ash, crude fiber, crude fat, crud e protein, and nitrogen-free extracts of the dietary treatments prepared in varying amounts of cocoyam meal.

APA 7th Edition Format for in-text citation and list of references.